

Quotient Space

Def. Let $(X, \mathcal{C})$ be a Topological space, and Y be a set.

For an onto map $f: X \rightarrow Y$, define a Quotient Topology on Y induced by f as

$$\mathcal{C}_\alpha \stackrel{\text{def}}{=} \{U \subseteq Y \mid f^{-1}[U] \text{ is open of } (X, \mathcal{C})\}.$$

It is actually a Topology on Y , because: clearly, $\emptyset, Y \in \mathcal{C}_\alpha$ and

$$U_\alpha \in \mathcal{C}_\alpha \ (\alpha \in I) \Rightarrow f^{-1}\left[\bigcup_{\alpha \in I} U_\alpha\right] = \bigcup_{\alpha \in I} f^{-1}[U_\alpha] \text{ is also open in } X, \text{ so is } \bigcup U_\alpha.$$

Similarly, $f^{-1}\left[\bigcap_{\alpha \in I} U_\alpha\right] = \bigcap_{\alpha \in I} f^{-1}[U_\alpha]$ also open if I is finite, so is $\bigcap U_\alpha$.

Moreover, it is the finest Topology on Y such that f is Continuous, because

$f: X \rightarrow Y$ is continuous iff $\forall U \in \mathcal{C}_Y, f^{-1}[U] \in \mathcal{C}_X$ iff $\mathcal{C}_Y \subseteq \mathcal{C}_\alpha$.

where $\mathcal{C}_Y$ is arbitrary Topology on Y . $\square$

Def. Let X be a Topological Space, and $\sim$ be an equivalence relation on X .

Define the Canonical map on X as

$$\pi: X \rightarrow X/\sim; x \mapsto [x]$$

The $(X/\sim, \mathcal{C}_\pi)$ is called Quotient Space where $\mathcal{C}_\pi$ is a Quotient topology on $X/\sim$, induced by π .

Let X be a Topological Space, $\sim$ be an equivalence relation on X , π be a Canonical map.
 Lemma. Let Z be a Topological space and a map $g: X/\sim \rightarrow Z$,

g is Continuous if and only if $g \circ \pi$ is Continuous,
 $\downarrow X \rightarrow Z$

Proof. Since $\pi: X \rightarrow X/\sim$ is Continuous, trivially $g \circ \pi$ is Continuous.

Conversely, assume $g \circ \pi$ is Continuous. Then, given an open $U \subseteq Z$,

$(g \circ \pi)^{-1}[U] = \pi^{-1}[g^{-1}[U]]$ is open of X , so $g^{-1}[U]$ is open of $X/\sim$. $\square$

Lemma. Let Z be a Topological space.

If a Continuous map $f: X \rightarrow Z$ satisfies $x \sim y \Rightarrow f(x) = f(y)$,
 then a map induced by f :

$$\bar{f}: X/\sim \rightarrow Z: [x] \mapsto f(x)$$

is Continuous and it is a unique map such that $\bar{f} \circ \pi = f$.

Proof. $\bar{f}$ is well-defined by: $[x] = [y] \Leftrightarrow x \sim y \Rightarrow f(x) = f(y)$.

Given $x \in X$, $f(x) = \bar{f}([x]) = \bar{f}(\pi(x)) = (\bar{f} \circ \pi)(x)$, so $\bar{f} \circ \pi = f$

By the Lemma above, $\bar{f}$ is Continuous. And, given a map $g: X/\sim \rightarrow Z$ s.t. $g \circ \pi = f$,

$$g([x]) = g(\pi(x)) = f(x) = \bar{f}([x])$$

So the uniqueness is guaranteed.

Lemma. Let Z be a Topological space.

If a Continuous onto map $f: X \rightarrow Z$ satisfies $x \sim y \Leftrightarrow f(x) = f(y)$ and either open or closed,
 then a map induced by f :

$$\bar{f}: X/\sim \rightarrow Z: [x] \mapsto f(x)$$

is Homeomorphism.

Quotient map.

Def. Given Topological space X and Y , a map $f: X \rightarrow Y$ is called a quotient map if:
 f is Continuous, Onto, and satisfies for any $U \subseteq Y$,

$$U \text{ is open in } Y \stackrel{\text{f continuous}}{\Leftrightarrow} f^{-1}[U] \text{ is open in } X.$$

Quotient

Prop. Composition of Quotient map is also Quotient map.

Proof Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be quotient maps. Given open set $U \subseteq Z$,

$$(g \circ f)^{-1}[U] \text{ is open } \Leftrightarrow f^{-1}[g^{-1}[U]] \text{ is open } \Leftrightarrow g^{-1}[U] \text{ is open } \Leftrightarrow U \text{ is open}$$

And, $(g \circ f)[X] = g[f[X]] = g[Y] = Z$, $(g \circ f)$ is a Quotient map. $\square$

Prop. If a Continuous Onto map $f: X \rightarrow Y$ is either open or closed map, then f is Quotient map.

Proof Given $U \subseteq Y$, if $f^{-1}[U]$ is open, then $f[f^{-1}[U]] = U$ is open, if f is open map.

And, if $f^{-1}[U]$ is closed, then $f[X \setminus f^{-1}[U]]$ is closed if f is closed map.

Since f is onto, $X \setminus f^{-1}[U] = f^{-1}[Y \setminus U] = f^{-1}[Y \setminus U]$, so

$$f[X \setminus f^{-1}[U]] = f[f^{-1}[Y \setminus U]] = Y \setminus U \text{ is closed, or } U \text{ is open. } \square$$

But, there is a quotient map neither open nor closed: define $X = \{(x, y) \in \mathbb{R}^2 \mid x \geq 0 \text{ or } y = 0\}$.

$$f: X \rightarrow \mathbb{R}; (x, y) \mapsto x$$

Then, it is clearly onto, and Continuous is given by: for any $a < b$ in $\mathbb{R}$,

$$f^{-1}[(a, b)] = [(a, b) \times \mathbb{R}] \cap X \text{ is open.}$$

To show f is quotient map, suppose that $f^{-1}[U]$ is open for some $U \subseteq \mathbb{R}$.

Given $x \in U$, $f((x, 0)) = x \in U$, so $(x, 0) \in f^{-1}[U]$. Since $f^{-1}[U]$ is open, for some $\epsilon > 0$, $(x, 0) \in B_\epsilon((x, 0)) \cap X \subseteq f^{-1}[U]$. Thus, $f[B_\epsilon((x, 0))] = (x - \epsilon, x + \epsilon) \cap f[X] = (x - \epsilon, x + \epsilon) \subseteq U$.
 so, U is open.

Moreover, $B_1((0, 2)) \cap X$ is open of X , but $f[B_1((0, 2)) \cap X] = [0, 1)$, not open.

And, $F = \{(\frac{1}{n}, n) \mid n \in \mathbb{N}\}$ is closed because $F' = \emptyset$ so $F = \bar{F}$.

But, $f[F] = \{\frac{1}{n} \mid n \in \mathbb{N}\}$ is not closed in $\mathbb{R}$, since $0 \notin f[F]$ but $0 \in f[F]'$.

Comment: A projection $\pi: \mathbb{R}^2 \rightarrow \mathbb{R}$ cannot be closed map: $F = \{(\frac{1}{n}, n) \mid n \in \mathbb{N}\}$.